YUN-TING CHENG

ycheng3@caltech.edu | https://yuntingcheng.github.io/ | M.C. 367-17, 1200 E California Blvd, Pasadena, CA 91125
US Legal Permanent Resident (Green Card holder), Taiwan Citizenship

PROFESSIONAL EXPERIENCE

California Institute of Technology
Research Scientist — 2024 – present

Jet Propulsion Laboratory/California Institute of Technology
Postdoctoral Researcher — 2021 – 2024

EDUCATION

California Institute of Technology
Ph.D. in Physics (Advisor: Prof. Jamie Bock) — 2021
Thesis: Cosmology and Astrophysics with Intensity Mapping
M.S. in Physics — 2019

National Taiwan University
B.S. in Physics — 2014

RESEARCH INTERESTS

- Large-scale structure surveys
- Photometric redshift inference and calibration
- Line intensity mapping
- Extragalactic background

RESEARCH EXPERIENCE

California Institute of Technology / Jet Propulsion Laboratory — Pasadena, CA

Research Scientist — 2024 – present

- SPHEREx Mission
  - Characterizing the photometric redshift estimation and systematics
  - Optimizing the multi-tracer analysis for SPHEREx cosmology science
  - Modeling the extragalactic background and line intensity mapping signal for SPHEREx
- Photometric Redshift with Clustering Statistics (collaborators: E. Huff, K. Markovic, I. Szapudi)
  - Developing a novel photometric redshift technique with clustering information

Postdoctoral Researcher (Advisor: Dr. Olivier Doré) — 2021 – 2024

- 3D Light Cones Cosmology (collaborators: B. Wandelt, O. Doré, T.-C. Chang)
  - Developing a Bayesian data-driven technique to optimally constrain cosmology with spectral imaging data
- Galactic Extinction Modeling (collaborators: B. Hensley, O. Doré, T.-C. Chang)
  - Building Galactic dust extinction model from multi-tracer datasets
- Extragalactic Radio Dipole (collaborators: A. Lidz, T.-C. Chang)
  - Modeling the cosmic dipole signal from extragalactic radio sources
- Intra-halo Light Signal in the Extragalactic Background (collaborator: J. J. Bock)
  - Modeling non-linear clustering and intra-halo light in the near-infrared background

Graduate Research Assistant (Advisor: Prof. Jamie Bock) — 2015 – 2021

- Line Intensity Mapping
  - Developing analysis algorithms to tackle line confusion in line intensity mapping surveys
  - Establishing formalism of optimal mapping strategies for large-scale structure surveys
  - Modeling galaxy-intensity mapping cross-correlation for SPHEREx
- CIBER (Cosmic Infrared Background Experiment)
  - Building CIBER analysis pipeline and characterizing observational systematic effects
  - Studying intra-halo light with stacking analysis on CIBER images
- TIME (Tomographic Ionized Carbon Intensity Mapping Experiment)
  - Simulating signal and foregrounds for TIME analysis pipeline
  - Developing foreground mitigation techniques
  - Analyzing TIME instrument data
  - Participating in instrument deployment at the ARO 12m telescope

Academia Sinica of Astronomy and Astrophysics (ASIAA) Taipei, Taiwan
Research Assistant (Advisor: Tzu-Ching Chang) 2014 – 2015
· Developing foreground mitigation techniques for line intensity mapping

Summer Student (Advisors: Sheng-Yuan Liu, Yu-Nung Su, I-Ta Hsieh) 2013 – 2013
· Modeling the starless core with radiative transfer

GRANTS

- **NASA Astrophysics Data Analysis Program 24 (ADAP 24)** 2025 – 2027
  Title: Detecting Baryonic Acoustic Oscillations in the WISE galaxy sample
  Role: Co-I (PI: Katarina Markovic, JPL)
- **NASA Astrophysics Data Analysis Program 24 (ADAP 24)** 2025 – 2027
  Title: Cosmology with Juno: Extracting the Cosmic Microwave Background Dipole at Low Frequencies and the Nature of the Diffuse Radio Background
  Role: Co-I (PI: Tzu-Ching Chang, JPL)

TECHNICAL SKILLS

- **Statistical Tools:** Bayesian statistics, Markov Chain Monte Carlo, Fisher analysis, sparse reconstruction, convex optimization, Machine Learning
- **Programming Languages:** Python (JAX, Astropy, emcee, Pandas, scikit-learn, TensorFlow, Keras, seaborn), SQL, IDL, Matlab, C++, Fortran, LaTeX
- **Instrumentation:** SOLIDWORKS, trained in machine shop techniques

STUDENT MENTORING

- Blanca A Crazzolara, ETH Zurich, JPL Visiting Master Student Mar 2025 – present
  Topic: Photometric Redshift Calibration with Clustering
- Jean Choppin de Janvry, Paris-Saclay University, JPL Visiting Master Student Sep 2024 – Mar 2025
  Topic: Photometric Redshift for SPHEREx
- Gemma (Zhaoyu) Huai, Caltech Physics Graduate Student Jun 2024 – present
  Topic: SPHEREx Cosmology
- Jui-Kuan Chan, National Taiwan University, Caltech SURF Program Jun 2024 – Aug 2024
  Topic: SPHEREx Photometric Redshift With SOM
- Kailai Wang, Cornell University, JPL SURF Program Jun 2023 – Aug 2023
  Topic: Bayesian Multi-line Intensity Mapping
- Abby Williams, NYU/Caltech, Caltech Post-baccalaureate Student Jun 2023 – present
  Topic: Small-scale Nonlinear Effects in Cross Correlations

SERVICE AND OUTREACH

- Line Intensity Meeting 2026, Organizer (100+ participants) Aug 2026
- The Chinese University of Hong Kong Study Tour Program, Lecturer Jul 2025
- Caltech Cosmology Journal Club, Co-organizer Sep 2022 – present
- 237th AAS Meeting, Oral Session Chair Jan 2021
- Gabrielino High School, CA, Physics In-Class Activities Leader Jan 2020 – Sep 2022
- 233rd AAS Meeting, Poster Judge Jan 2019
- Journal referee: *Astrophysical Journal (ApJ)*, *Astrophysical Journal Letters (ApJL)*, *Journal of Cosmology and Astroparticle Physics (JCAP)*, *Monthly Notices of the Royal Astronomical Society (MNRAS)*, *Astronomy & Astrophysics (A&A)*

AWARDS AND HONORS

- Balzan Cosmological Studies Travel Award 2022
- Taiwan-Caltech Ministry of Education Fellowship 2015 – 2019
- Dean's Award of College of Science, National Taiwan University 2014

PRESENTATIONS

Invited Conference Talks and Lectures

- KITP Program: CMB x LSS (video) Santa Barbara, CA, Feb, 2026
- Cosmoglobe Bayesian Cosmological Data Analysis Course Oslo, Norway, Sep, 2025
- Cosmology on the Steep Rise Workshop Sexten, Italy, Feb, 2025
- A new era with Line Intensity Mapping Workshop Heidelberg, Germany, Dec, 2024
- Line Intensity Mapping Workshop MPA, Garching, Germany, Apr, 2023

Invited Institution Colloquia and Seminars

- CITA Seminar Toronto, Canada, Jul, 2026
- Perimeter Institute Cosmology Seminar Waterloo, Canada, Jul, 2026
- Caltech IPAC Seminar Pasadena, CA, Oct, 2025
- ASIAA Colloquium Taipei, Taiwan, Dec, 2024
- University of Bonn/MPIfRA Colloquium Bonn, Germany, Dec 2024
- KIPAC Seminar Stanford, CA, Nov, 2024
- ASIAA Seminar Taipei, Taiwan, Oct, 2023
- LAM Cafe Club LAM, Marseille, France, Dec, 2022
- IAP Universe Seminar IAP, Paris, France, Nov, 2022
- NYU CCPP Seminar NYU, NY, Aug, 2022
- ICAP seminar (virtual) IAP, Paris, France, Jun, 2022
- ASIAA Seminar (video) Taipei, Taiwan, May, 2022
- IRSIG Webminar (virtual), Oct, 2021
- ASIAA Seminar (virtual) Taipei, Taiwan, Mar, 2021
- UChicago KICP Seminar (virtual) Chicago, IL, Jan, 2021
- Berkeley BCCP Seminar (virtual) Berkeley, CA, Dec, 2020
- CCA Flatiron Institute Cosmology Group Meeting (virtual) CCA, NY, Oct, 2020
- OSU CCAPP Seminar (virtual) OSU, OH, Oct, 2020
- JHU Cosmology/GW Journal Club (virtual) JHU, MD, Oct, 2020
- UPenn Astronomy Seminar (virtual) UPenn, PA, Sep, 2020
- CCA Flatiron Institute Lunch Talk (virtual) CCA, NY, Sep, 2020
- ASIAA Seminar Taipei, Taiwan, Sep, 2018

Contributed Talks

- Line Intensity Mapping 2026 Pasadena, CA, Aug, 2026
- Infrared Spectroscopy from Space Pasadena, CA, Oct, 2025
- Line Intensity Mapping 2025 Annecy, France, Jun, 2025
- PRIMA and the Future of Far-Infrared Science Pasadena, CAMay, 2025
- Roman Community Workshop (video) Pasadena, CAJul, 2024
- Line Intensity Mapping 2024 Workshop (video) UIUC, IL, Jun, 2024
- Diffuse Cosmic Backgrounds and the Low Surface Brightness Universe Aspen, CO, Apr, 2024
- Caltech ObsCos Seminar Pasadena, CAFeb, 2024
- 243rd AAS Meeting New Orleans, LA, Jan, 2024
- Pan-Ex Meeting (virtual), Nov, 2023
- JPL Postdoc Seminar JPL, Pasadena, CA, Nov, 2023
- Probing the Universe at High Resolution Conference Taipei, Taiwan, Nov, 2023
- LAM CONCERTO Working Group Meeting LAM, Marseille, France, Dec, 2022
- Caltech ObsCos Seminar Pasadena, CAOct, 2022
- Columbia Cosmology Group Seminar Columbia, NY, Aug, 2022
- Cosmology from Home (video) (virtual), Jun, 2022
- Cross Correlations with CHORD Workshop (virtual), Oct, 2021
- SUBLIME Workshop (virtual), Oct, 2021
- KICP Line Intensity Mapping Workshop (virtual) Chicago, IL, Jul, 2021

- Caltech ObsCos Seminar (virtual) Pasadena, CAFeb, 2021
- 237th AAS Meeting (virtual), Jan, 2021
- Caltech ObsCos Seminar (virtual) Pasadena, CASep, 2020
- CCAT-prime Science Working Group Meeting (virtual) Cornell, NY, Sep, 2020
- Caltech ObsCos Seminar Pasadena, CAFeb, 2020
- L2S2 : Lines in the Large Scale Structure Conference Marseille, France, Jul, 2019
- Caltech ObsCos Seminar Pasadena, CAJun, 2019
- Caltech ObsCos Seminar Pasadena, CAMay, 2019
- 233rd AAS Meeting Seattle, WA, Jan, 2019
- Taiwanese Theoretical Astrophysics Workshop Taipei, Taiwan, Sep, 2018
- Caltech ObsCos Seminar Pasadena, CAJun, 2018
- Cosmological Signals from Cosmic Dawn to the Present Aspen, CO, Feb, 2018
- Caltech ObsCos Seminar Pasadena, CADec, 2017
- Caltech ObsCos Seminar Pasadena, CANov, 2016
- Caltech ObsCos Seminar Pasadena, CAJun, 2016
- Opportunities and Challenges in Intensity Mapping Workshop KIPAC, CA, Mar, 2016
- ASROC Annual Meeting (Taiwanese Astronomical Society) Ilan, Taiwan, May, 2015

PUBLICATIONS

See ADS, Google Scholar, and INSPIRE for the complete publication list

First-author (11 published / accepted)

1. *Feature Intensity Mapping: Polycyclic Aromatic Hydrocarbon Emission from All Galaxies Across Cosmic Time*
Y.-T. Cheng, B. S. Hensley, T. S.-Y. Lai, 2025, ApJ, 993, 30, ADS, [arXiv:2506.13863]
2. *Mapping Galactic Dust Emission and Extinction with HI, HII, and H_2*
Y.-T. Cheng, B. S. Hensley, T.-C. Chang, & O. Doré, 2025, ApJ, 985, 15, ADS, [arXiv:2411.12801]
3. *Bayesian Multi-line Intensity Mapping*
Y.-T. Cheng, K. Wang, B. D. Wandelt, T.-C. Chang, & O. Doré, 2024, ApJ, 971, 159, ADS, [arXiv:2403.19740]
4. *Is the Radio Source Dipole from NVSS Consistent with the CMB and ΛCDM?*
Y.-T. Cheng, T.-C. Chang, & A. Lidz, 2024, ApJ, 965, 32, ADS, [arXiv:2309.02490]
5. *Data-driven Cosmology from Three-dimensional Light Cones*
Y.-T. Cheng, B. D. Wandelt, T.-C. Chang, & O. Doré, 2023, ApJ, 944, 151, ADS, [arXiv:2210.10052]
6. *Near-infrared Extragalactic Background Light Fluctuations on Nonlinear Scales*
Y.-T. Cheng & J. J. Bock, 2022, ApJ, 940, 115, ADS, [arXiv:2207.13712]
7. *Cosmic Near-Infrared Background Tomography with SPHEREx Using Galaxy Cross-Correlations*
Y.-T. Cheng & T.-C. Chang, 2022, ApJ, 925, 136, ADS, [arXiv:2109.10914]
8. *Probing Intra-Halo Light with Galaxy Stacking in CIBER Images*
Y.-T. Cheng et al. (CIBER Collaboration), 2021, ApJ, 919, 69, ADS, [arXiv:2103.03882]
9. *Phase-Space Spectral Line De-confusion in Intensity Mapping*
Y.-T. Cheng, T.-C. Chang, & J. J. Bock, 2020, ApJ, 901, 142, ADS, [arXiv:2005.05341]
10. *Optimally Mapping Large-Scale Structures with Luminous Sources*
Y.-T. Cheng, R. de Putter, T.-C. Chang, & O. Doré, 2019, ApJ, 877, 86, ADS, [arXiv:1809.06384]
11. *Spectral Line De-Confusion in an Intensity Mapping Survey*
Y.-T. Cheng, T.-C. Chang, J. J. Bock, C. M. Bradford, & A. R. Cooray, 2016, ApJ, 832, 165, ADS, [arXiv:1604.07833]

Contributing author

1. *Comparing the Near-infrared Spectral Energy Distributions from Different Stellar Population Synthesis Models with SPHEREx Observations*
J. Lee, et al., 2026, submitted, ADS, [arXiv:2607.18048]
2. *A UV-to-Near-infrared QSO Composite Spectrum from the SPHEREx All-Sky Survey*
M. Kim, et al., 2026, submitted, ADS, [arXiv:2607.10282]
3. *TIME Commissioning Observations: II. On-sky Characterization and the 2D Map Data Processing Pipeline*
B. Vaughan, et al., 2026, submitted, ADS, [arXiv:2607.01220]
4. *The SPHEREx View of Galaxy Clusters: A Simulation-based Validation of the Forced Photometry Pipeline for Extended Sources*
H. Banhk, et al., 2026, submitted, ADS, [arXiv:2606.19875]
5. *SPHEREx mapping of diffuse PAH and H II emission in the Galactic plane*
G. Murgia, et al., 2026, ApJ, 1004, 176, ADS, [arXiv:2603.23620]
6. *The SPHEREx Instrument: Calibration, testing and performance measurements of the NIR 2 spectroscopic surveyor from the laboratory to in-orbit commissioning*
P. M. Korngut, et al., 2026, submitted, ADS, [arXiv:2603.29835]
7. *The SPHEREx Ices Investigation: An Overview*
G. J. Melnick, et al., 2026, ApJ, accepted, ADS, [arXiv:2603.22135]
8. *Spectral response of SPHEREx*
H. Hui, et al., 2026, submitted, ADS, [arXiv:2602.09139]

9. *Euclid preparation. Decomposing components of the extragalactic background light using multi-band intensity mapping cross-correlations*
Y. Cao, et al., 2026, submitted, ADS, [arXiv:2601.21111]

10. *SPHEREx Pre-Perihelion Mapping of H2O, CO2, and CO in Interstellar Object 3I/ATLAS*
C. M. Lisse, et al., 2025, ApJL, 1000L, 52, ADS, [arXiv:2512.07318]

11. *TIME Commissioning Observations: I. Mapping Dust and Molecular Gas in the Sgr A Molecular Cloud Complex at the Galactic Center*
S. F. Yang, et al., 2026, ApJ, 1003, 23, ADS, [arXiv:2511.09473]

12. *The Redshifts from 122 Bands: Comparative Redshift Forecast for Low-Resolution Spectra from SPHEREx and 7-Dimensional Sky Survey (7DS)*
J. Bae, et al., 2025, A&A, 706, A347 ADS, [arXiv:2512.24537]

13. *The SPHEREx Satellite Mission*
Bock, et al., 2025, ApJ, 999, 139, ADS, [arXiv:2511.02985]

14. *Simulating Spectral Confusion in SPHEREx Photometry and Redshifts*
Z. Huai, et al., 2025, ApJ, 1000, 56, ADS, [arXiv:2510.01410]

15. *The SPHEREx Sky Simulator: Science Data Modeling for the First All-Sky Near-Infrared Spectral Survey*
B. P. Crill, et al., 2025, ApJS, 281, 10, ADS, [arXiv:2505.24856]

16. *The Potential of the SPHEREx Mission for Characterizing PAH 3.3 μm Emission in Nearby Galaxies*
E. Zhang, A. L. Faisst, et al., 2025, ApJ 992, 32, ADS, [arXiv:2503.21876]

17. *CIBER 4th flight fluctuation analysis: Measurements of near-IR auto- and cross-power spectra on arcminute to sub-degree scales*
R. M. Feder, et al., 2025, ApJ, 994, 30, ADS, [arXiv:2501.17933]

18. *CIBER 4th flight fluctuation analysis: Pseudo-power spectrum formalism, improved source masking and validation on mocks*
R. M. Feder, et al., 2025, ApJ, 991, 87, ADS, [arXiv:2501.17932]

19. *Inferred Measurements of the Zodiacal Light Absolute Intensity through Fraunhofer Absorption Line Spectroscopy with CIBER*
P. M. Korngut, et al., 2022, ApJ, 926, 133, ADS, [arXiv:2104.07104]

20. *Probing Cosmic Reionization and Molecular Gas Growth with TIME*
G. Sun, T.-C. Chang, et al., 2021, ApJ, 915, 33, ADS, [arXiv:2012.09160]

21. *Superresolution Reconstruction of Severely Undersampled Point-spread Functions Using Point-source Stacking and Deconvolution*
T. Symons, M. Zemcov, et al., 2021, ApJS, 252, 24, ADS, [arXiv:2102.01094]

22. *Hafnium Films and Magnetic Shielding for TIME, A mm-Wavelength Spectrometer Array*
J. Hunacek, et al., 2018, JLTP, 193, 893, ADS

23. *A Foreground Masking Strategy for [C II] Intensity Mapping Experiments Using Galaxies Selected by Stellar Mass and Redshift*
G. Sun, L. Moncelsi, M. P. Viero, et al., 2018, ApJ, 856, 107, ADS, [arXiv:1601.10095]

24. *Design and fabrication of TES detector modules for the TIME-Pilot [C II] Intensity Mapping Experiment*
J. Hunacek, et al., 2016, JLTP, 184, 733, ADS

White paper / review paper contributions

1. *PRIMA General Observer Science Book Volume 2*
A. Moullet, et al., 2025, ADS, [arXiv:2511.10927]
Case #17 – Probing PAH from All Galaxies Across Redshifts – PAH Intensity Mapping with PRIMA [ADS]

2. *PRIMA General Observer Science Book*
A. Moullet, et al., 2023, ADS, [arXiv:2310.20572]
Case #9: Probing the history of cosmic star formation, black hole growth, and metallicity/dust evolution with line-intensity mapping

3. *Tomography of the Cosmic Dawn and Reionization Eras with Multiple Tracers*
T.-C. Chang, et al., 2019, Astro2020 White Paper, ADS, [arXiv:1903.11744]

4. *Line-Intensity Mapping: 2017 Status Report*
E. D. Kovetz, M. P. Viero, et al., 2017, ADS, [arXiv:1709.09066]

Conference proceedings

1. *A status update on TIME: a mm-wavelength spectrometer designed to probe the Epoch of Reionization*
A. Crites, et al., 2020, SPIE, 114530G

2. *Detector modules and spectrometers for the TIME-Pilot [CII] intensity mapping experiment*
J. Hunacek, et al., 2016, SPIE, 99140L